Microsoft

Session 1 – Copilot Studio Level 200

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Agenda

Copilot Studio 101 (45-60 min)

Email Assistant Overview & Architecture (30-45 min)

Break (20 min)

Hands-on Lab (>1 hour)

Wrap-up and Q&A (15 min)

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Copilot Studio 101

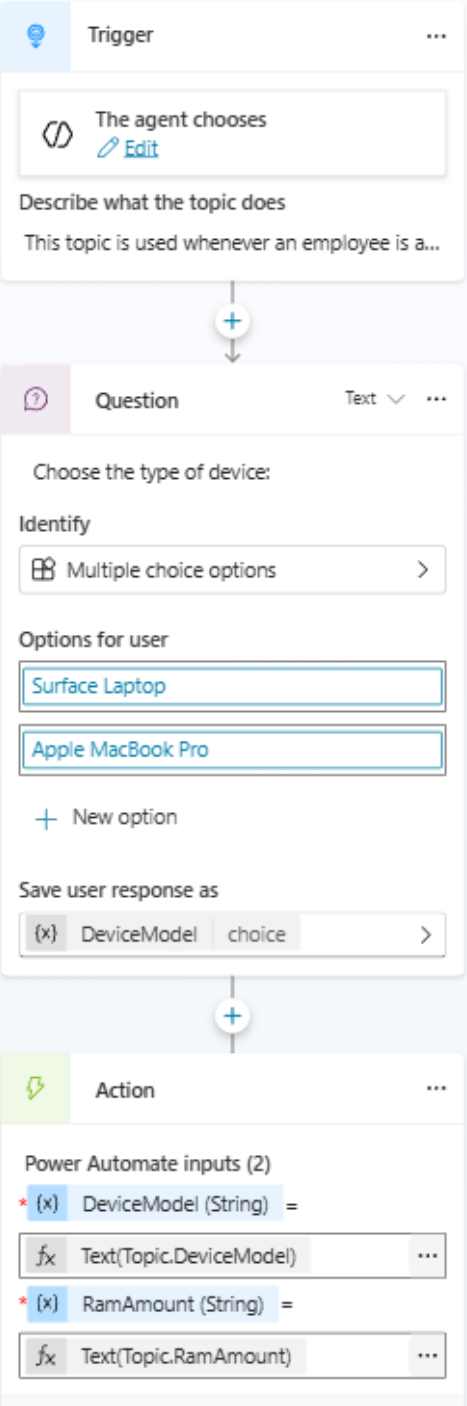
Before Generative AI

Everything was driven by **Topics**

Topics represent **specific and fixed** conversation paths within agent, and they manage the interactions between users and the AI.

Within a topic, you can:

- Ask questions to the user
- Call actions
- Use Adaptive Cards to provide a richer UI experience

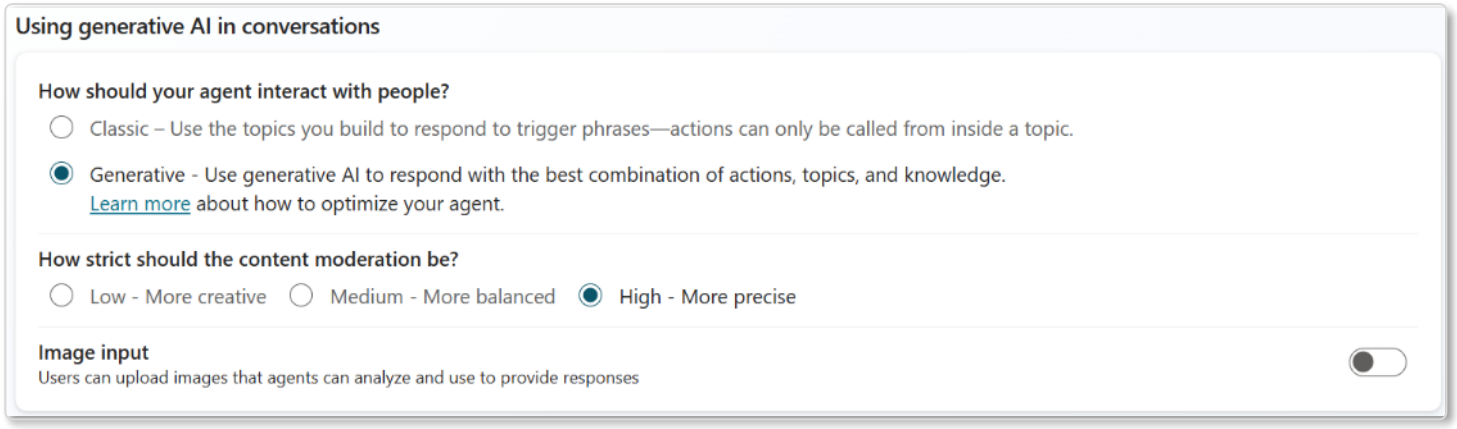


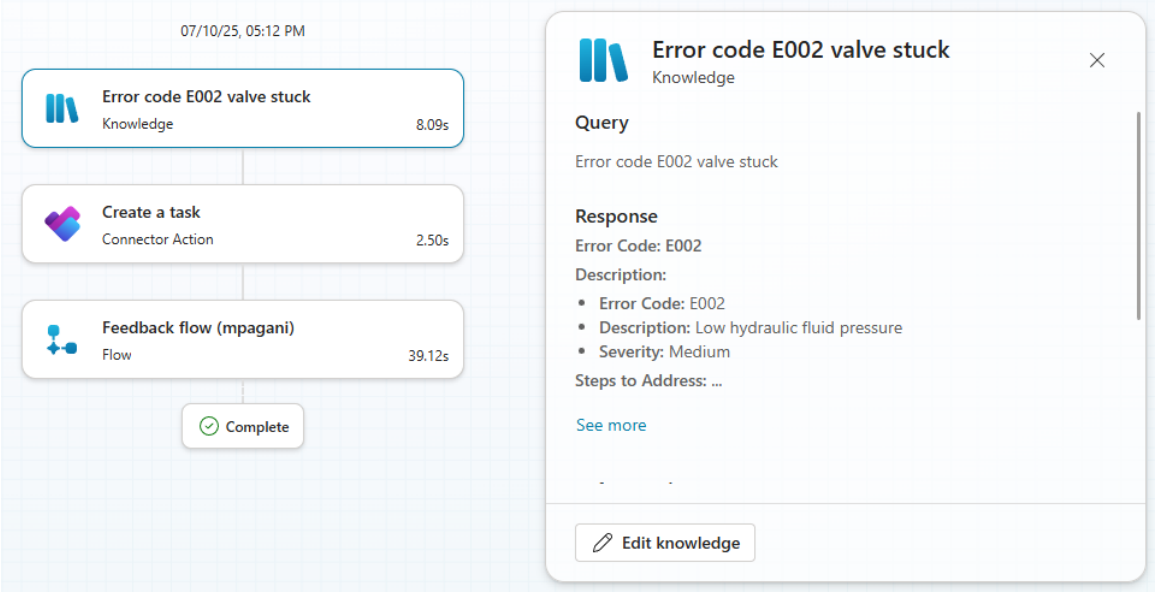
With Generative AI

When a user sends a message, your agent selects the most appropriate knowledge and / or more actions prepare its response. Multiple factors determine the selection. The most important factor is the **description** of the topics and actions. Other factors include the **name** of a topic or actions, any **input or output** parameters, and their names and descriptions.

If you selected **Generative** for how your agent should decide to respond, it's important to provide a **high-quality description** for each of the actions and topics in your agent, to ensure your agent selects the right topics and actions to respond to users.

The **Activity Map** let you see the decision flow of the agent.





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Add Knowledge sources

Supported Knowledge Sources

Name	Source	Description	Number of inputs supported in general answers	Authentication
Public Website	External	Searches the query input on Bing, only returns results from provided websites	4 public URLs (example, microsoft.com)	None
Documents	Internal	Searches documents uploaded to Dataverse, returns results from the document contents	Limited by Dataverse file storage allocation	None
SharePoint		Connects to a SharePoint URL, uses Graph Search to return results	4 URLs	Agent user's Microsoft Entra ID authentication *
Dataverse (Preview)		Connects to the connected Dataverse environment and uses retrieval-augmented generative technique in Dataverse to return results	Two Dataverse knowledge sources (and up to 15 tables per knowledge source)	Agent user's Microsoft Entra ID authentication *
Enterprise data via Copilot Connectors		Copilot Connectors enable organizations to bring external data from popular enterprise platforms (ServiceNow, Jira Cloud, Azure DevOps, etc.) into the tenant	Two per custom agent	Agent user's Microsoft Entra ID authentication *

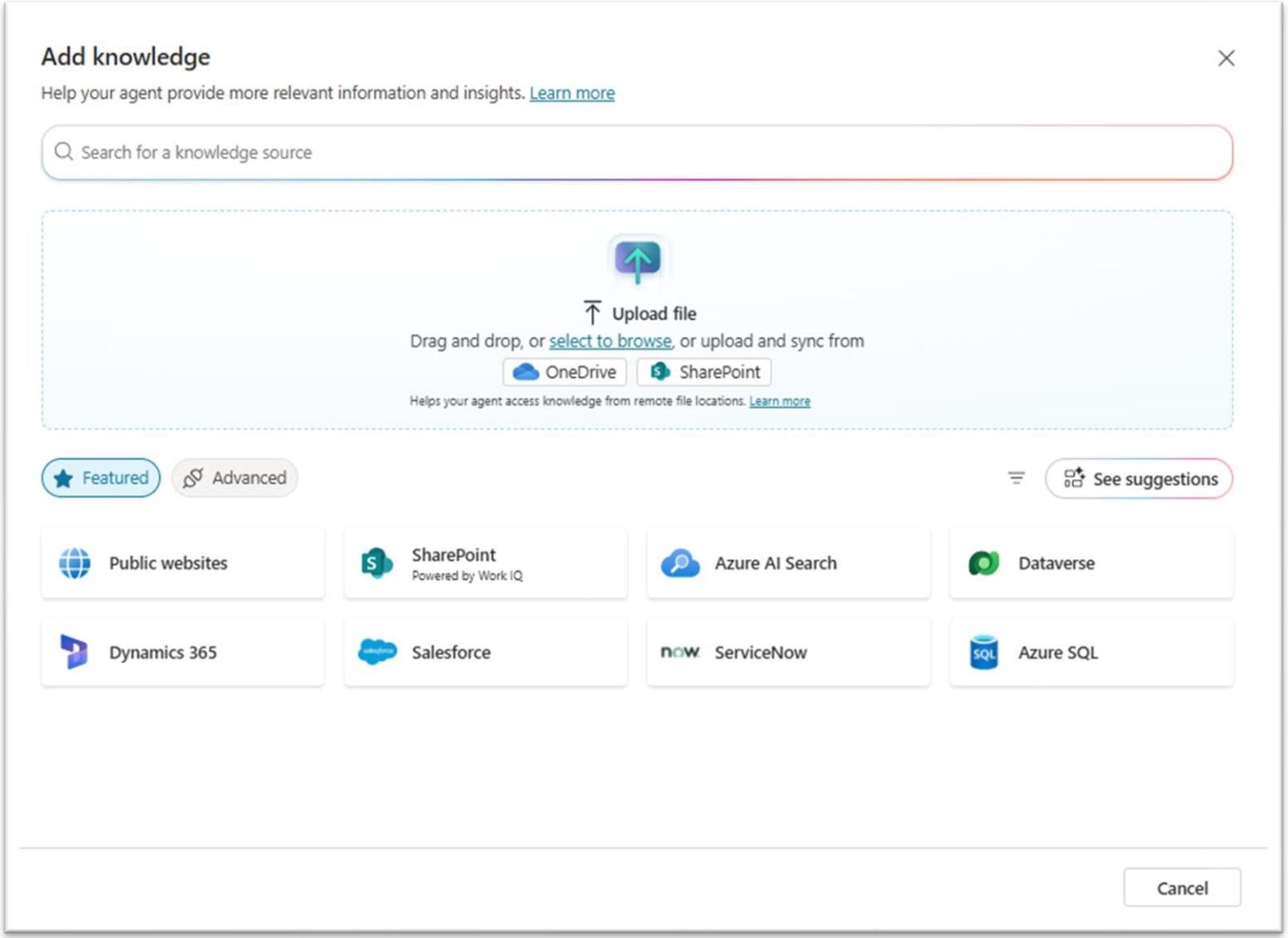
** Agent **user authentication** for information sources means that when a specific user asks a question of the agent, the agent only surfaces content for that specific user to access.*

Agent Knowledge

Provide appropriate description for knowledge sources for faster response

Configurable
-Agent General knowledge
-Tenant graph grounding with semantic search

Copilot connectors may require user authentication



Public website as Knowledge Source

Searches the query input on Bing, only returns results from provided websites.

Max 4 public URLs are supported.

URL Considerations:

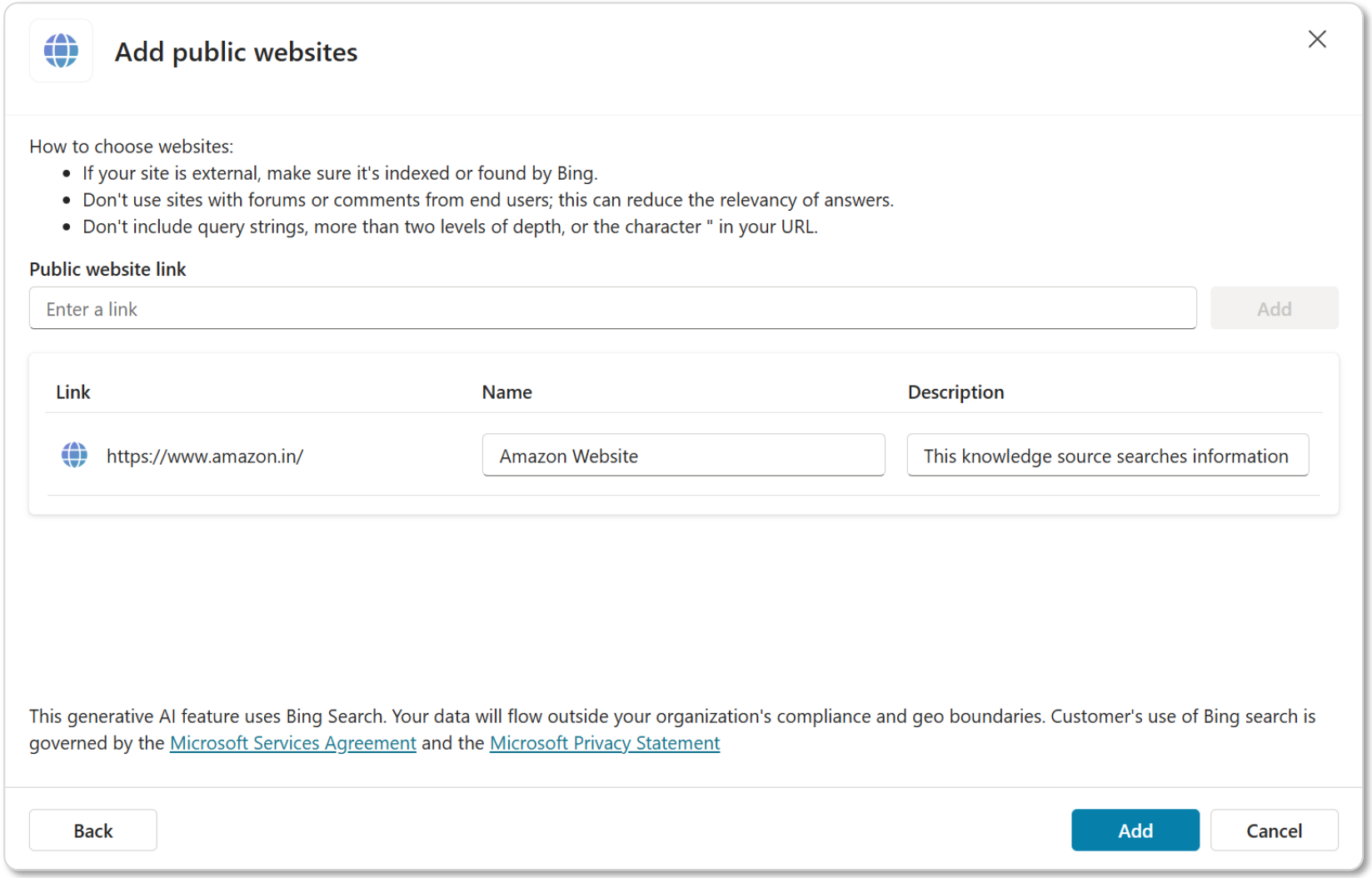
- The URL can have up to two levels of depth, also called "subpaths," indicated by forward slashes /.
- A trailing forward slash isn't included in the limit of two slashes.

Valid URL:

- www.contoso.com
- www.fabrikam.com/engines/rotary
- www.fabrikam.com/engines/rotary/

Not a valid URL:

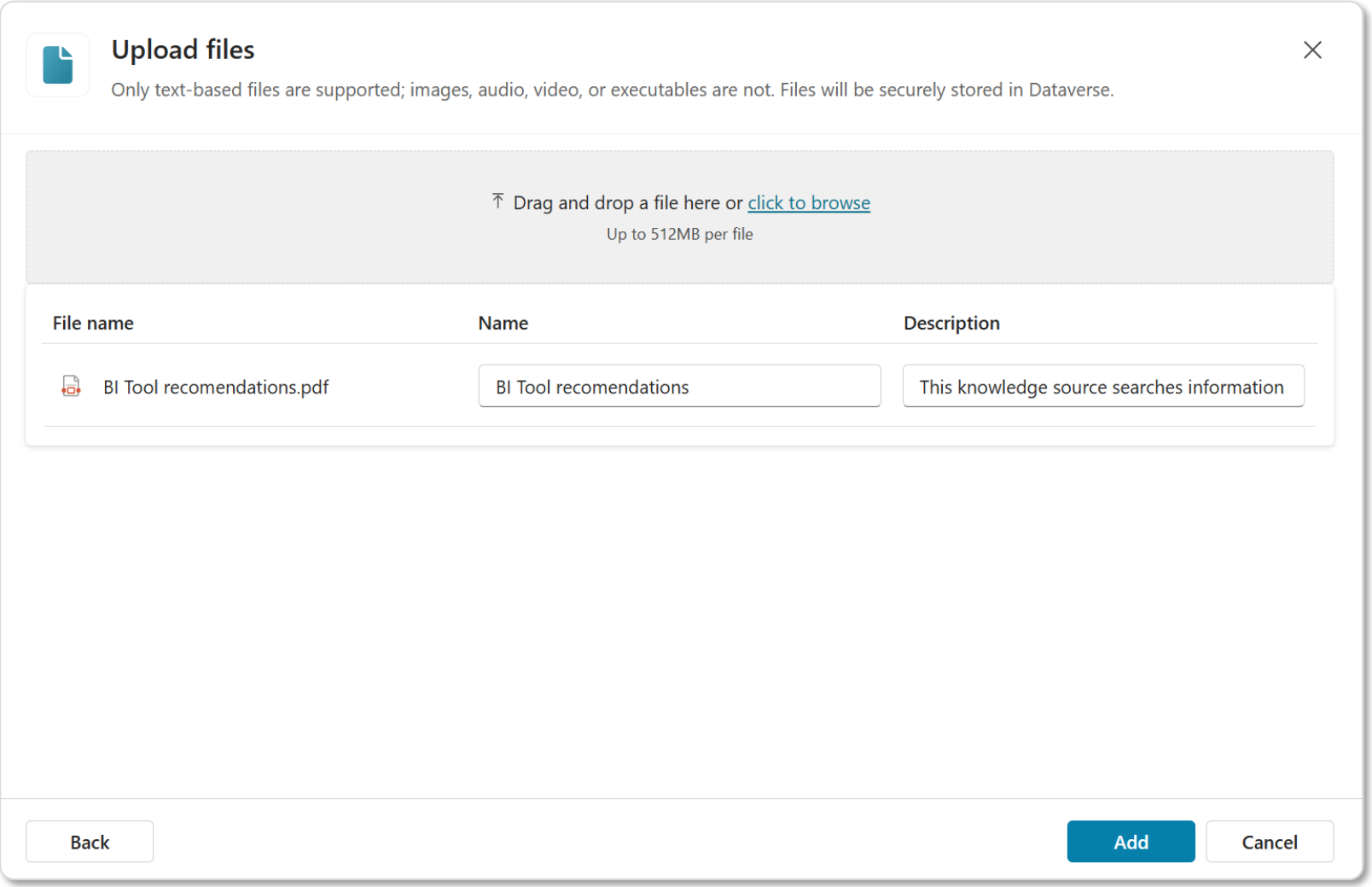
- www.fabrikam.com/engines/rotary/dual-shaft



Uploading file as Knowledge Source

The following file types are supported:

- Word documents (doc, docx)
- Excel spreadsheets (xls, xlsx)
- PowerPoint documents (ppt, pptx)
- PDF documents (pdf)
- Text documents (txt, md, log)
- HTML files (html, htm)
- CSV files (csv)
- XML files (xml)
- OpenDocument files (odt, ods, odp)
- EPUB documents (epub)
- Rich Text Format documents (rtf)
- Apple iWork documents (pages, key, numbers)
- JSON files (json)
- YAML files (yml, yaml)
- LaTeX files (tex)

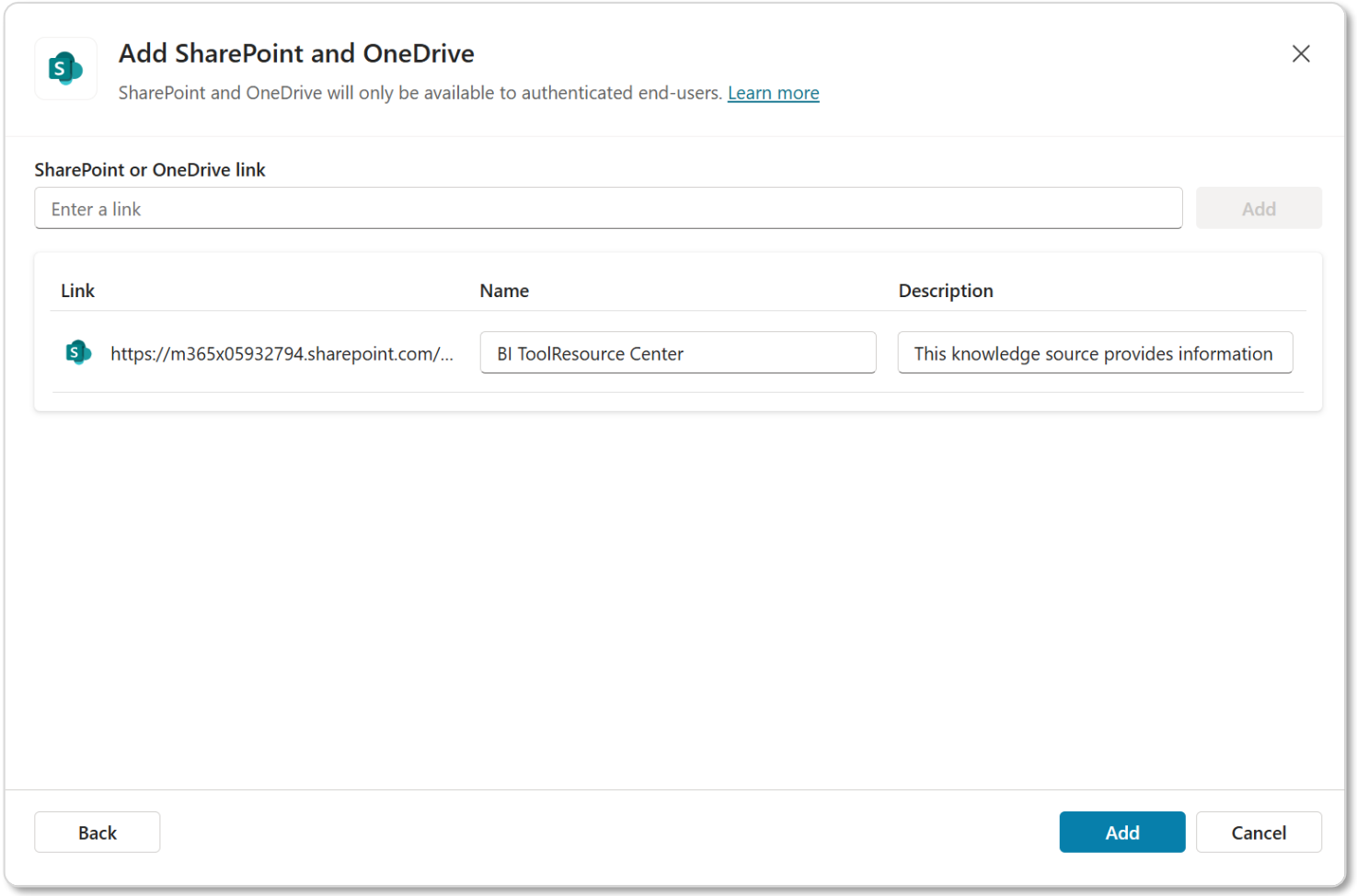


SharePoint as Knowledge Source

- Copilot Studio can leverage content from your internal SharePoint sites and libraries to respond to user queries.
- By pairing your agent with a SharePoint URL (e.g., contoso.sharepoint.com/sites/policies), agent searches the specified URL and its sub-paths for relevant information.

Best Practices/recommendations:

1. Max **4 URLs** are supported in classic orchestration, no limit for generative orchestration.
2. Agent **user's permissions** define which files can be retrieved. You will not see files if you don't have access to them.
3. Only **modern pages, Word, PowerPoint, PDF files** are supported.
4. Can handle files up to 7 Mb, files up to 200 Mb can be searched with **Enhanced Search Results** (requires M365 Copilot license)
5. Users need to **authenticate** to get AI generated content from SharePoint in both chats:
   - A test chat.
   - An agent's chat with a published agent.



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Instructions

Prompting practices

Let’s explore the following prompting practices:

- **Always / never / must** limitations
- **One shot / few shot** learning via examples
- **Conditional** instructions
- Formatting **cueing**
- **Chain-of-thought** split and explain step-by-step

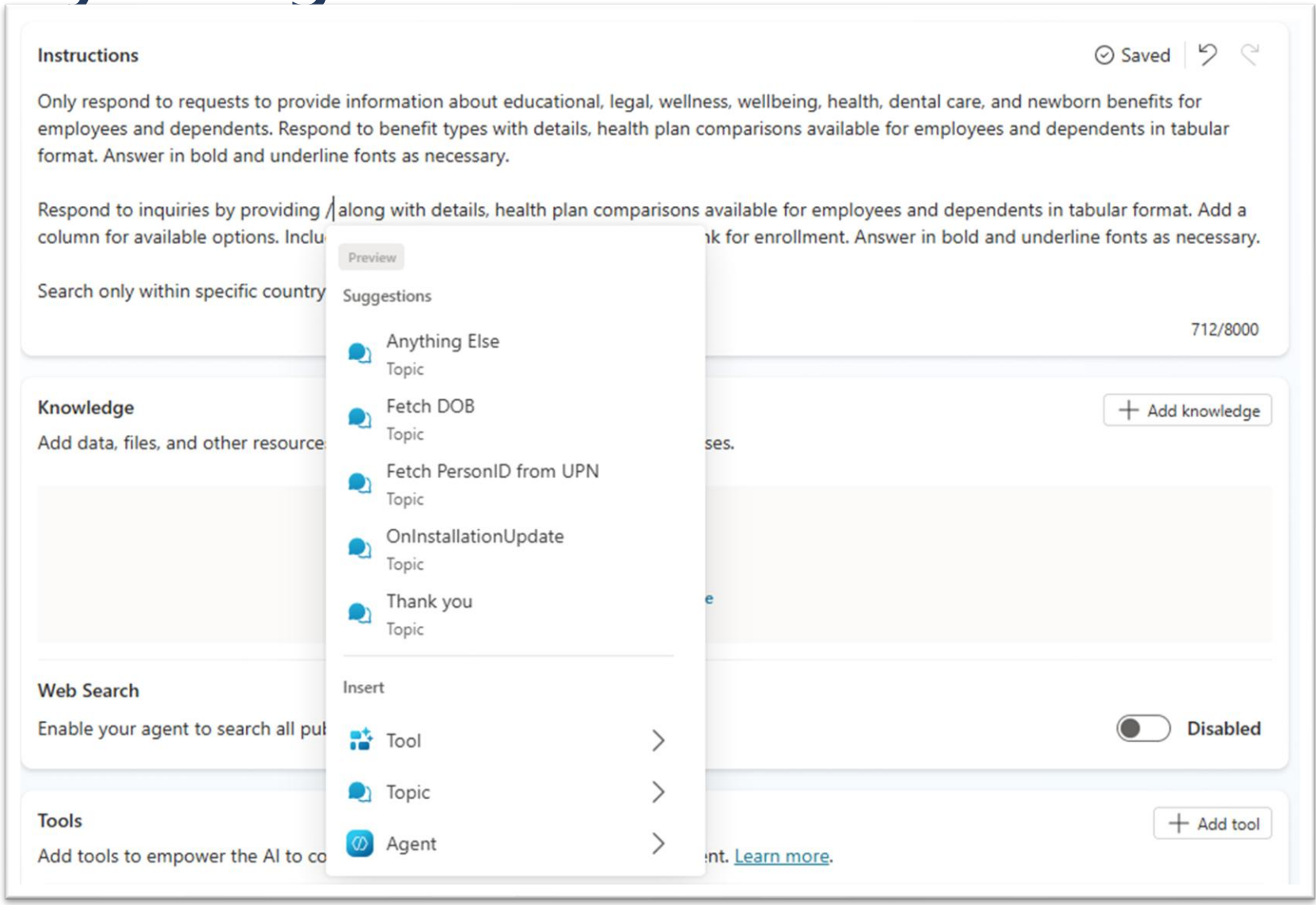
Instructions to craft your agent

Type instructions in plain text.

Within the instructions, type a slash (/) to add a reference to a specific resource, such as a tool, a topic, another agent, a variable, or a Power Fx expression.

Agent orchestrator determine which ones to call and when

Works when Generative orchestration is turned ON



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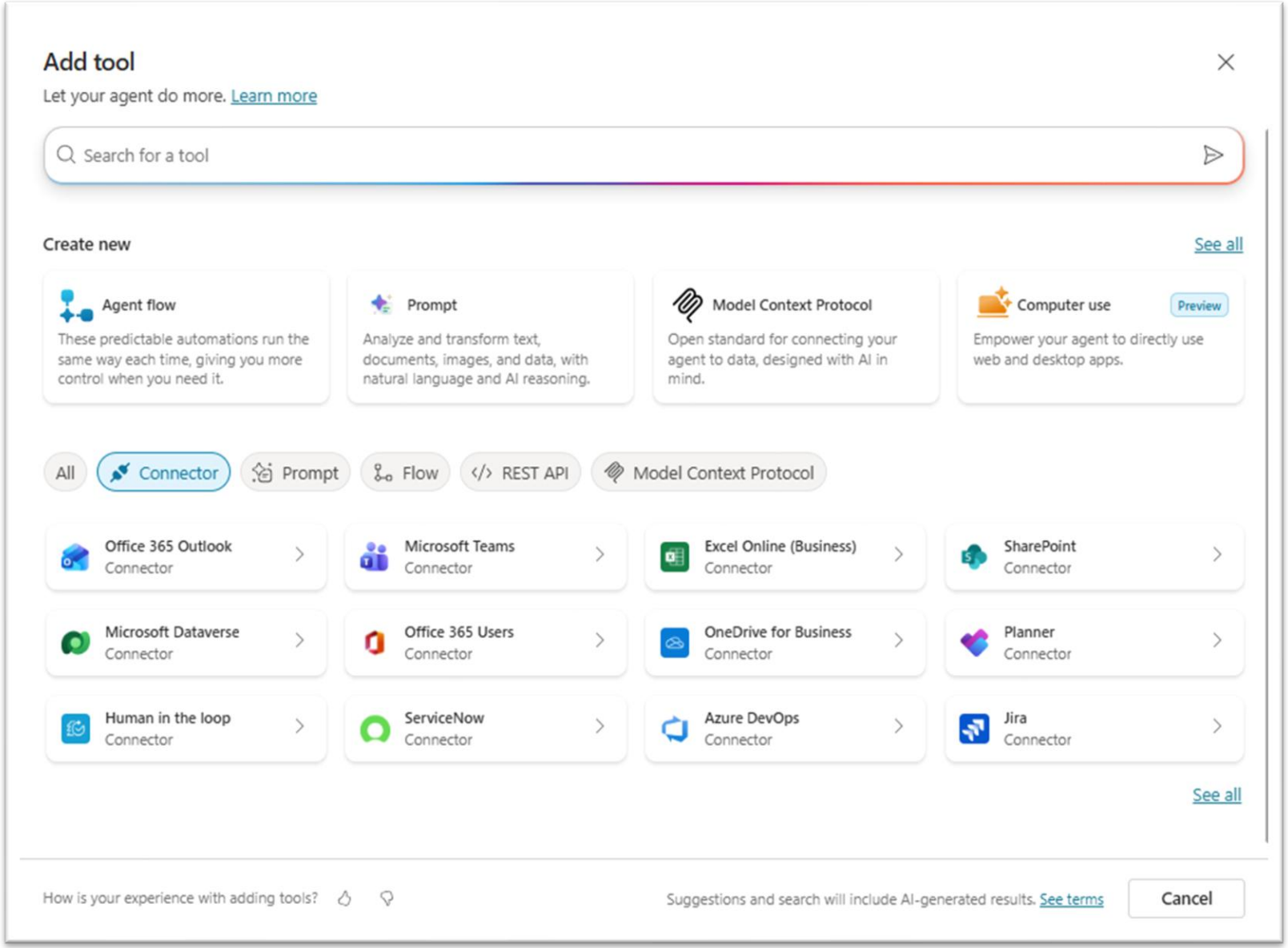
Tools

Copilot Studio Agent Tools

Let your agent interact with external systems

Perform various actions in response to user requests or autonomous triggers

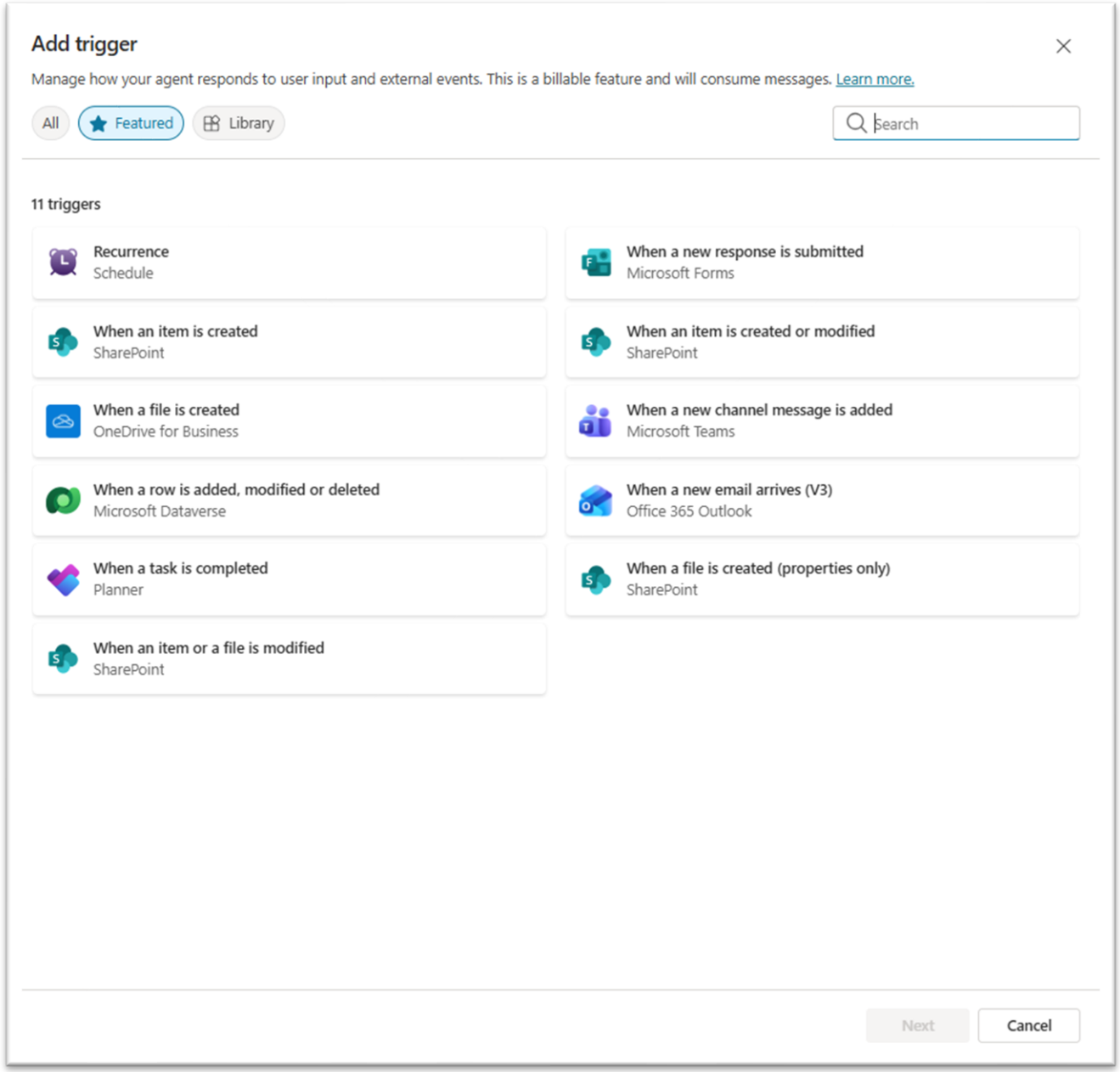
Agent can use tools to respond to users automatically, using [generative orchestration](#).
Tools can also be called explicitly from within a [topic](#)



Agent Triggers

Perform actions or call topics in response to an event

Event triggers allow agent to act autonomously in response to the occurring of defined event



Prompts

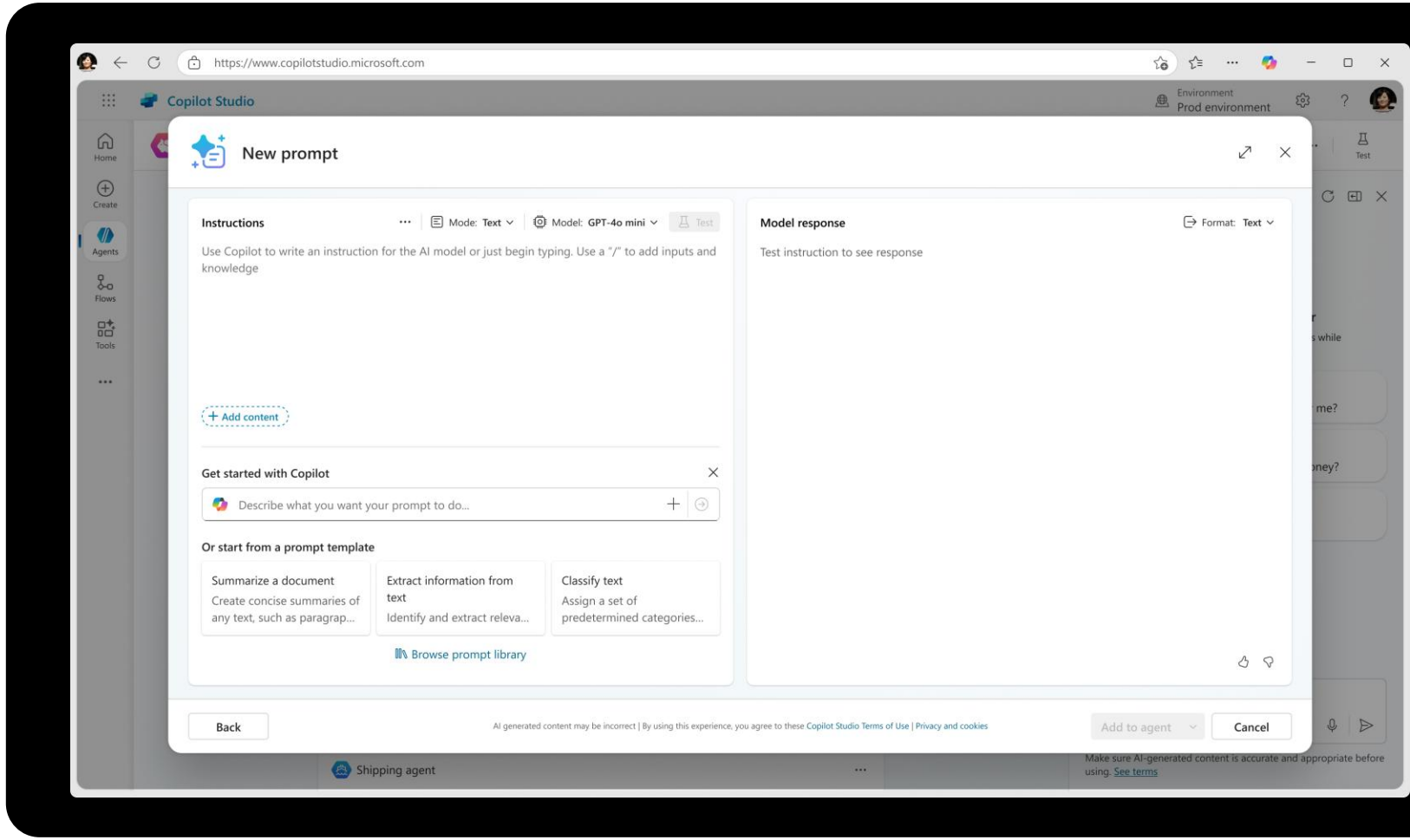
Provide structured instructions to guide the LLM to perform specific tasks

The Prompts tool allows makers to easily create **custom prompts** inside Copilot Studio, either at the app level or within an agent or topic.

Use out-of-the-box **templates** or write the prompt **from scratch**.

Add **deep reasoning** prompts to solve more complex business processes by integrating advanced models with your agents. *(Available in public preview)*.

Ground models in your enterprise data and securely draw insights from internal resources.



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Topics

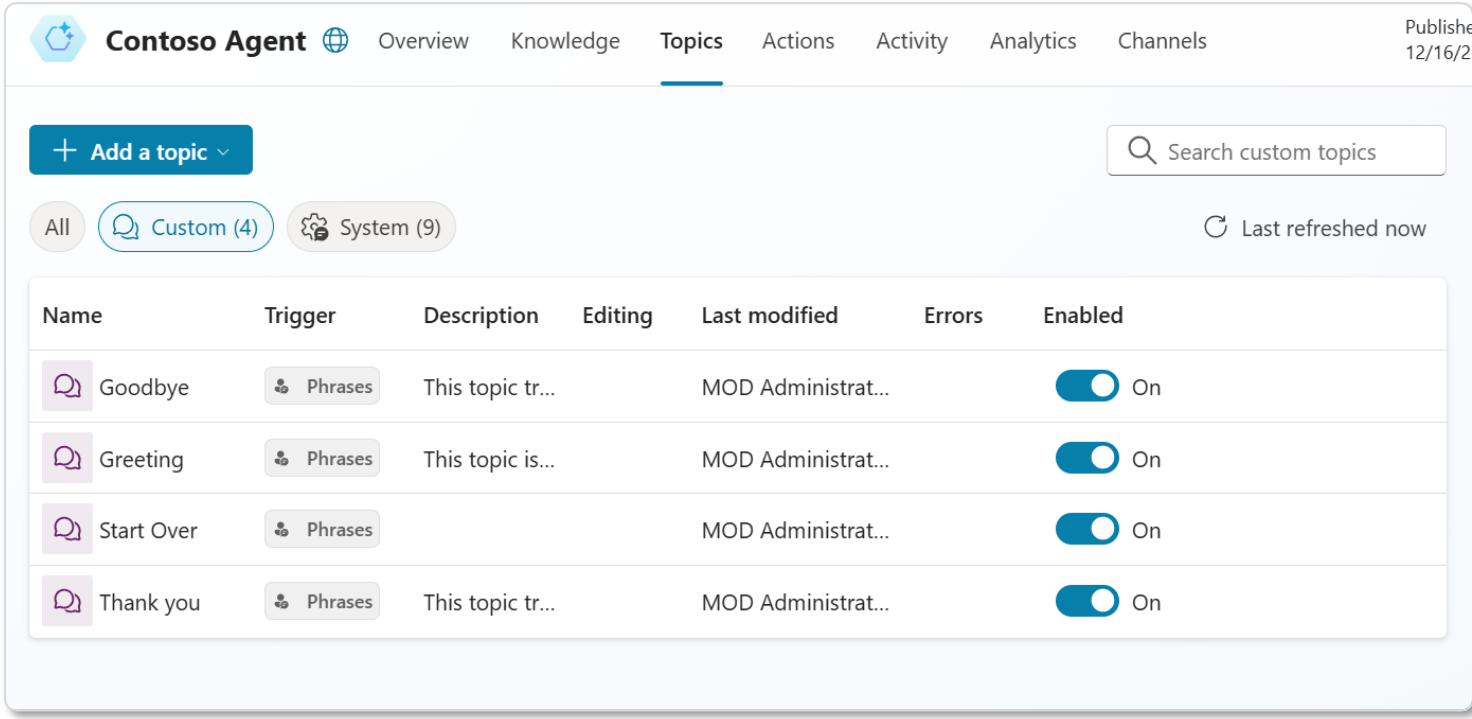
What are Topics in Copilot Studio?

A Topic defines how an agent conversation progresses.

Topics represent specific conversation paths within agent, allowing natural and seamless interactions between users and the AI.

These predefined paths guide the flow of conversation and enhance the overall experience

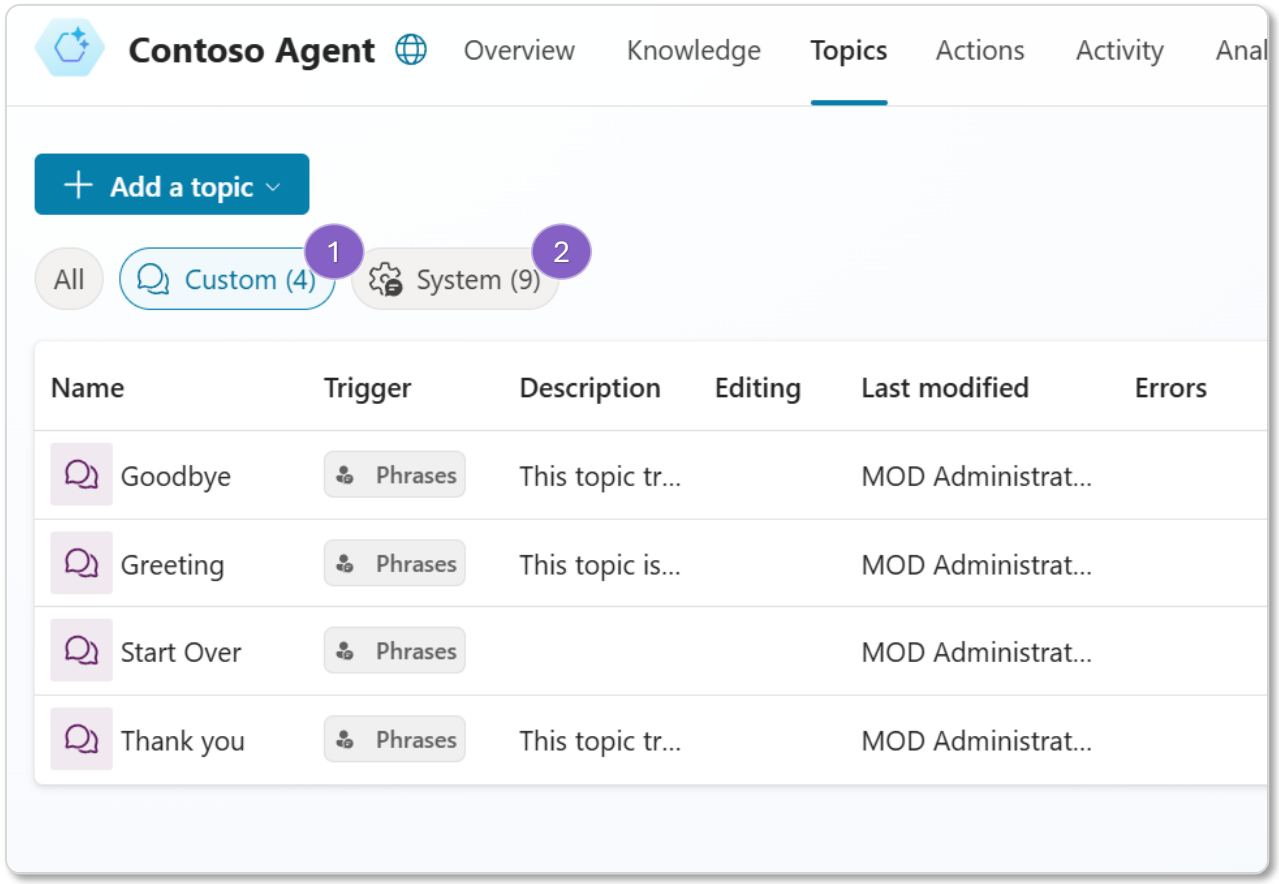
They were the core component of bots; they're less relevant with agents and the advent of Generative AI



Topic types

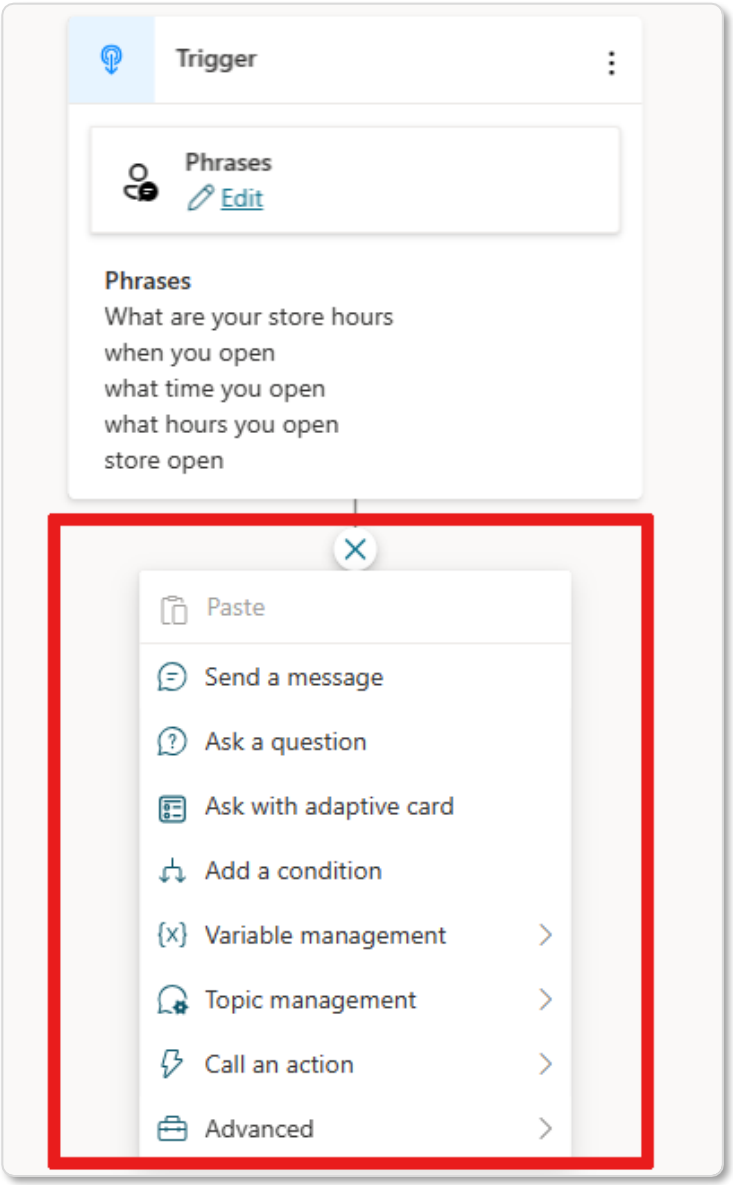
1. **Custom topic:**
   - **User-Defined**: Custom topics are created and defined by the user.
   - **Customizable**: Users can fully customize these topics, including the triggers and responses.
   - There are 3 lesion topics added for leaning purpose and recommended to disable for production agents.

2. **System topic:**
   - System topics in agent are made to trigger automatically based on relevant events.
   - You can also choose to start system topics manually by redirecting the conversation to them.
   - System topics cannot be deleted but can be disabled.



Conversational nodes

Option	Description
Send a message	Send the customer a message.
Ask a question	Ask the customer a question.
Ask with adaptive card	Create a JSON snippet to exchange with other apps.
Add a condition	Branch your conversation based on a condition.
Variable management	Set a value, parse a value, or clear all variables.
Topic management	Redirect, transfer, or end the topic or conversation.
Call an action	Call a flow like Power Automate or Excel Online or use a connector or a plugin.
Advanced	Generative answers, HTTP requests, events, and more.



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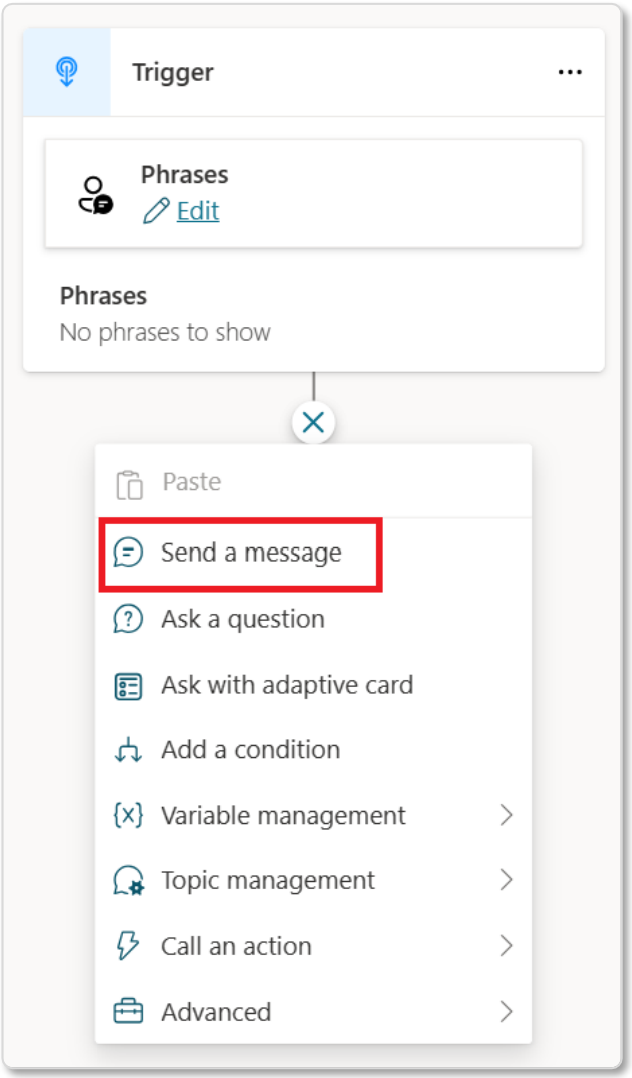
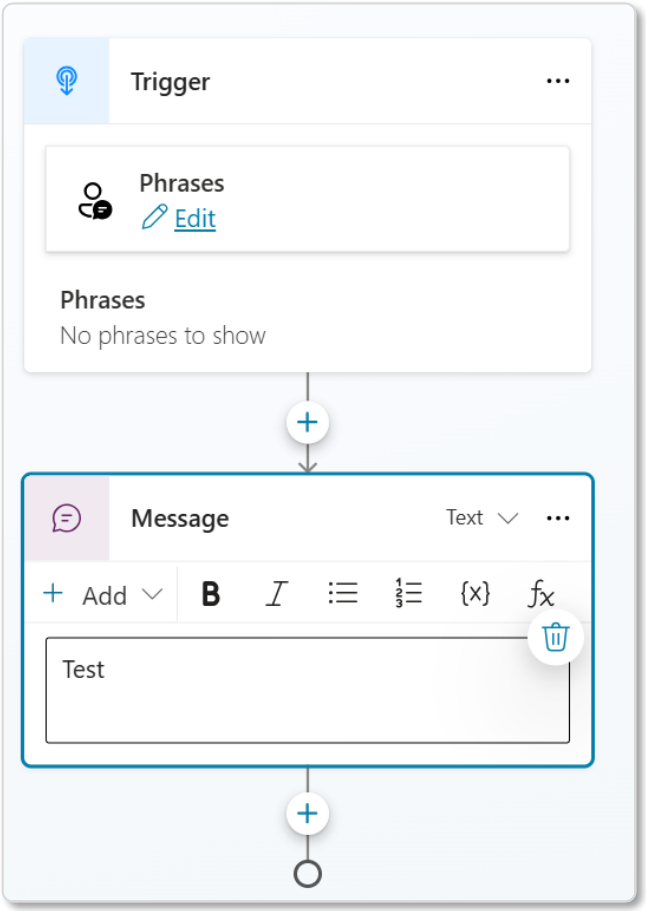
‘Send a message’ node

Message Node

The Message node, within the flow of a topic, sends a message from the agent to the user.

Messages can be simple text messages, but can also include richer components, such as images, videos, quick replies, and cards.





Message Types

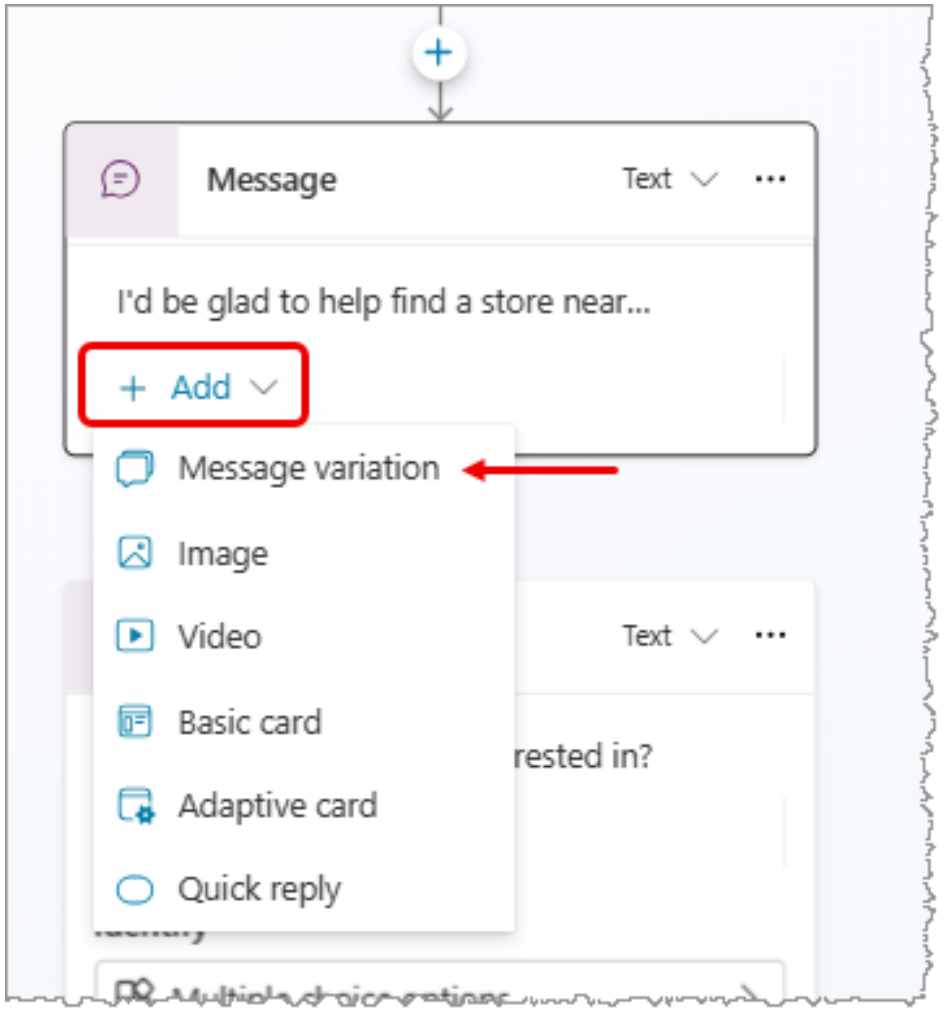
Message Variation - agent randomly picks one of them to use each time the node is triggered.

Image - To add an image, you need one hosted through a URL.

Video - Enter the URL of your video. The URL can either be a direct link to a publicly accessible MP4 file or a YouTube URL.

Adaptive card - Platform-agnostic cards that you can customize. You can design Adaptive Cards using the Adaptive Cards Designer or author the JSON directly.

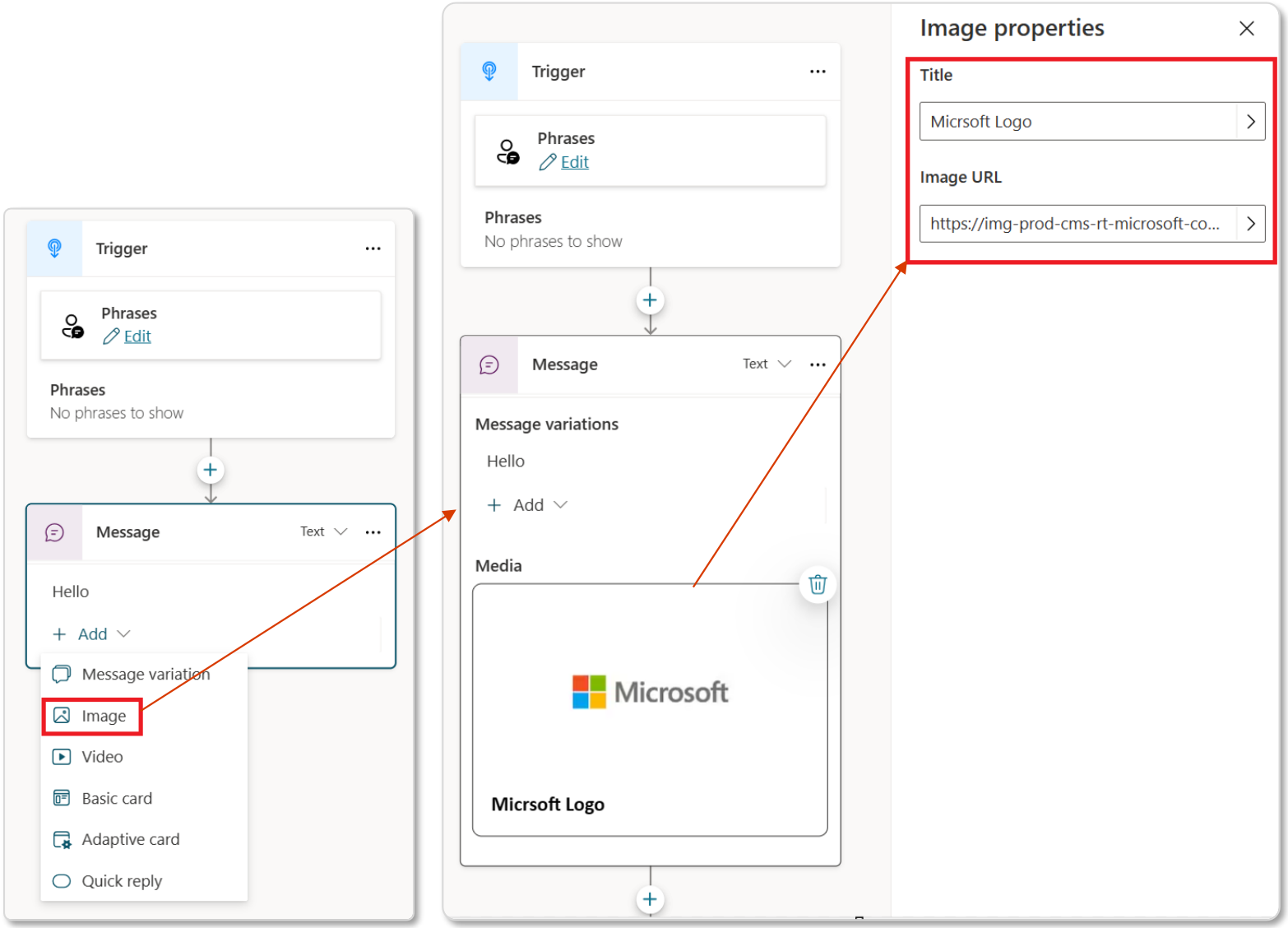
Quick reply - Quick replies suggest possible responses or actions for the user. Any action taken by the user is relayed back to the agent.



Add an image

To add an image, you need one hosted through a URL.

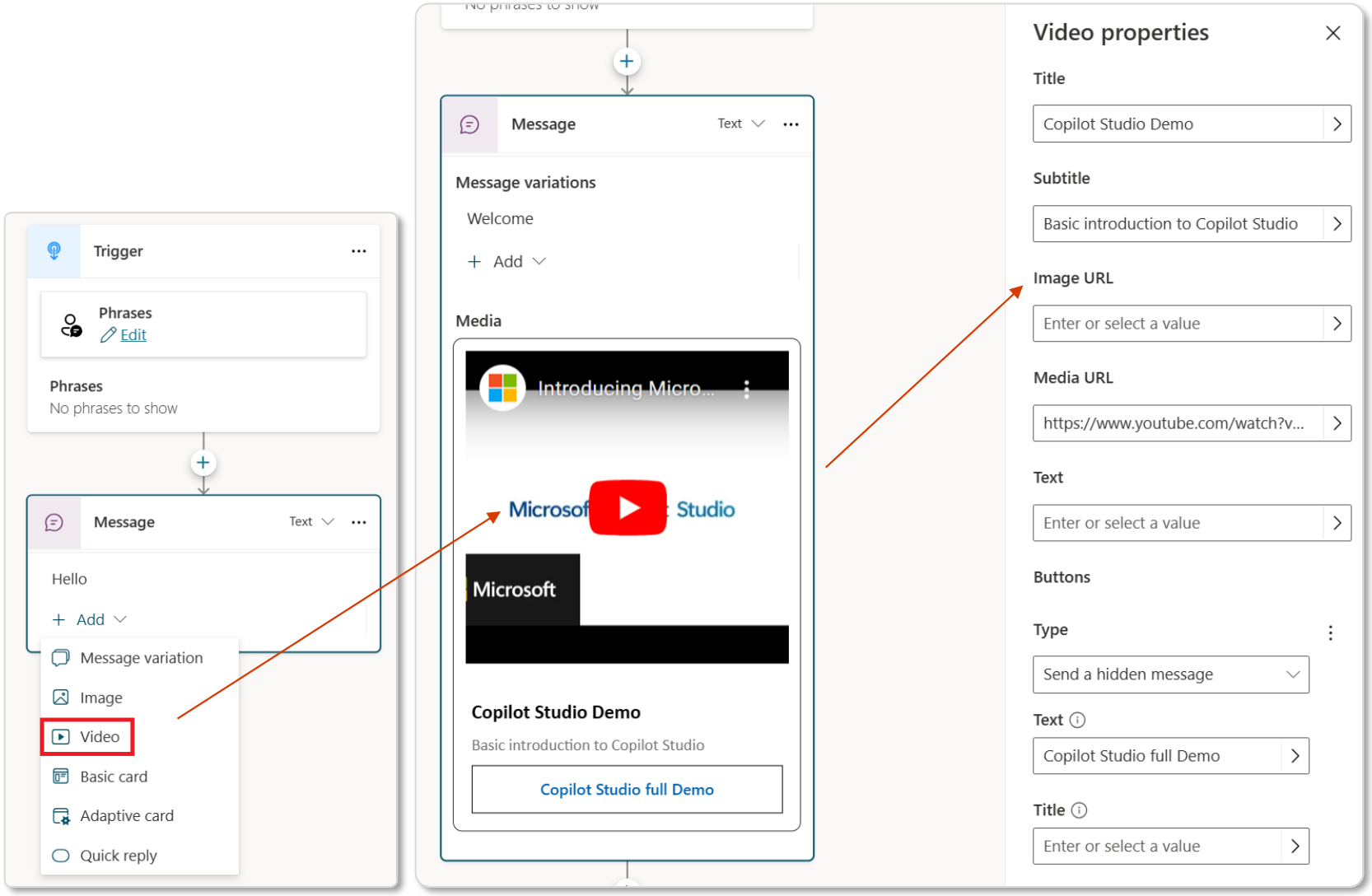
Title text will be displayed on the image.



Add a video

Under Media, enter the URL of your video. The URL can either be a direct link to a publicly accessible MP4 file or a YouTube URL.

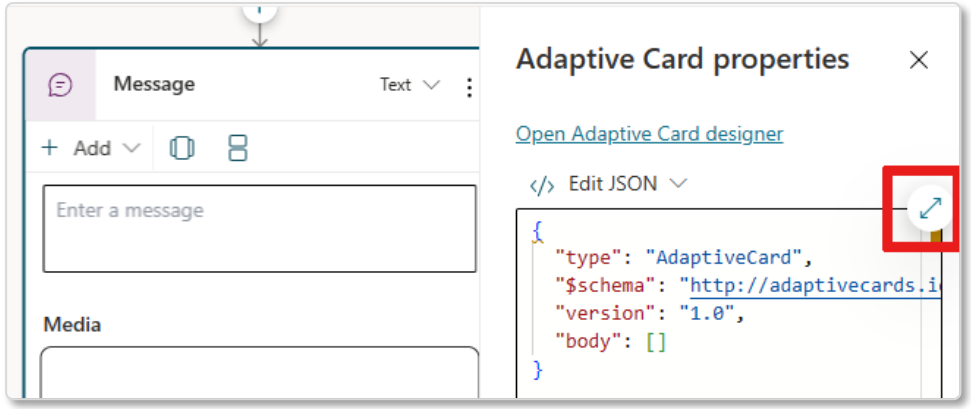
Optionally, enter a Title, Subtitle, Image (the URL of a publicly accessible image file), or Text to be shown alongside the video on the card.

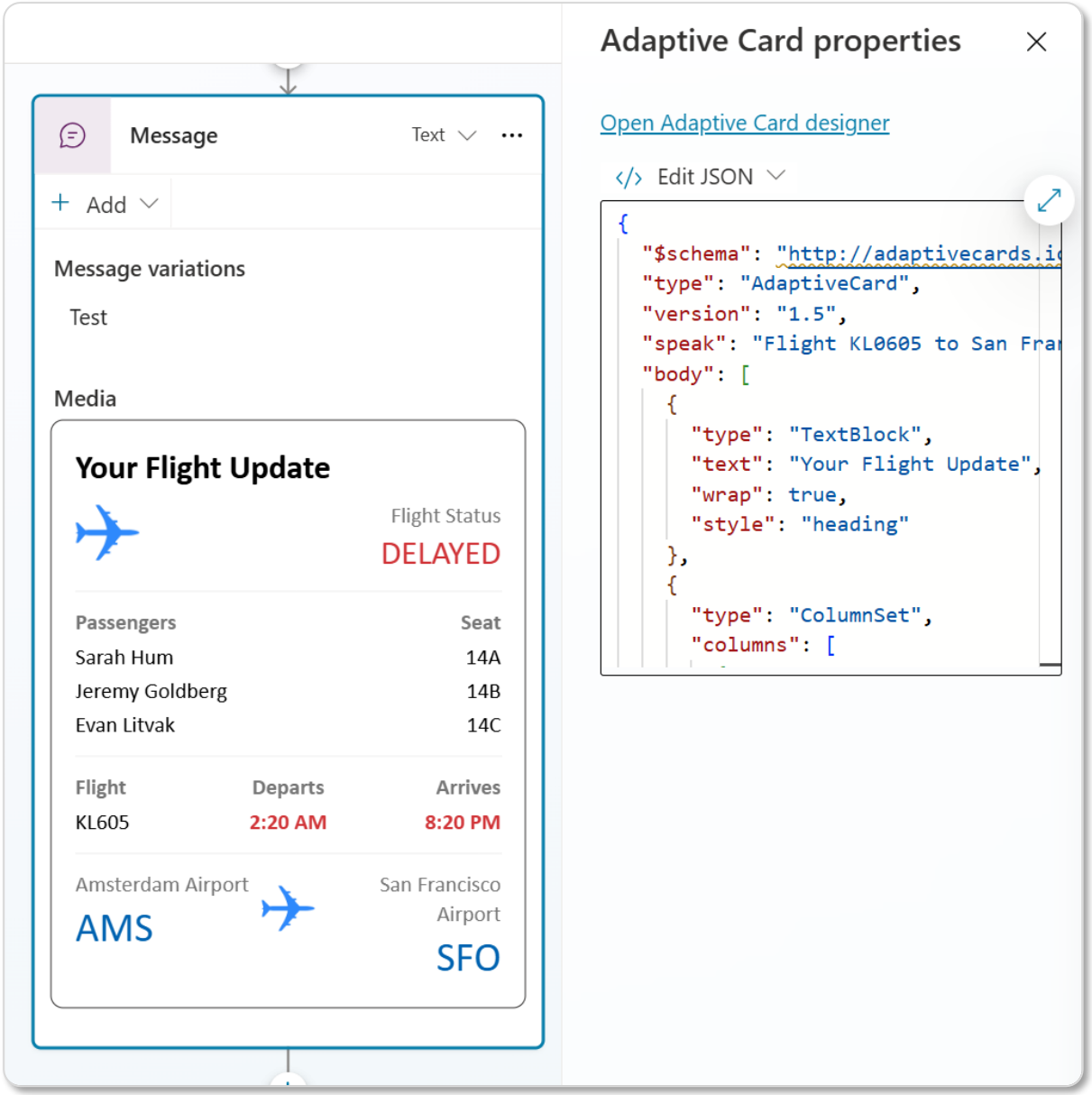


Adaptive card

Adaptive Cards are platform-agnostic cards that you can customize.

You can design Adaptive Cards using the Adaptive Cards Designer or author the JSON directly.





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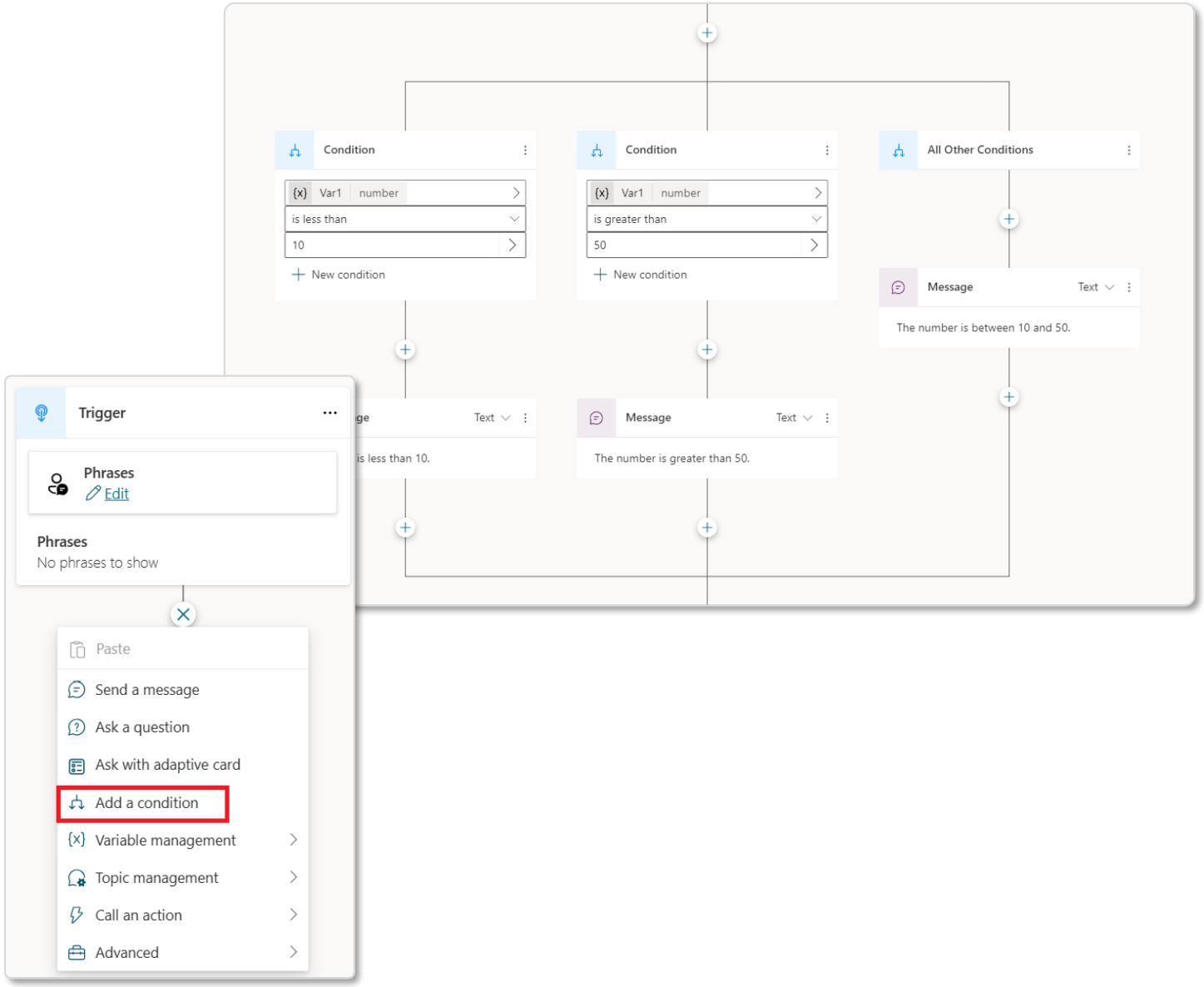
‘Add a condition’ node

Add a condition node

Conditions utilize branch logic to force the conversation path to branch based on specific criteria.

Example 1: Health Insurance Agent asks the user about their age and suggests policies based on age above or below 60 years.

Example 2: If the value of isClubMember is True, then the topic offers a discount; if the value is False, the topic presents the standard item price



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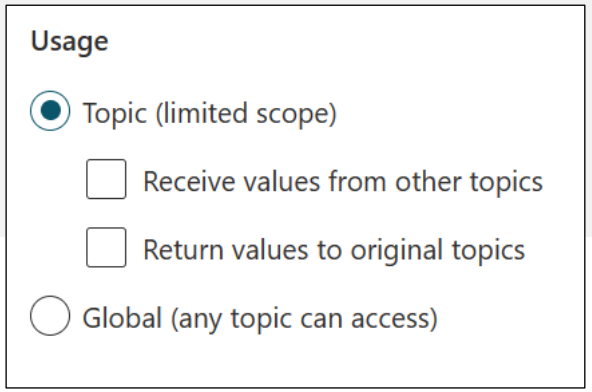
‘Variable management’ node

Variable scope

Use variables to save customers responses in an agent conversation and use them later in the conversation or pass them to other topics or Power Automate flows.

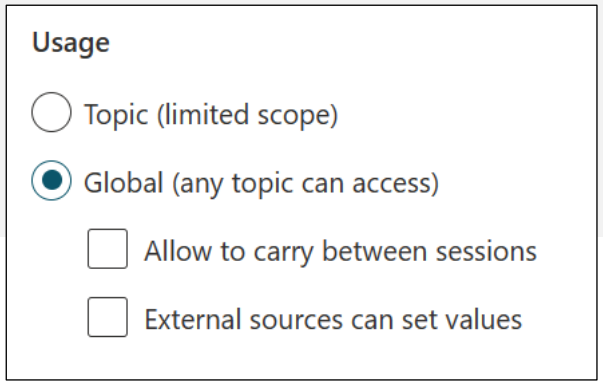
Topic variables

These are specific to the topics in which they are created and are the default type for newly created variables.



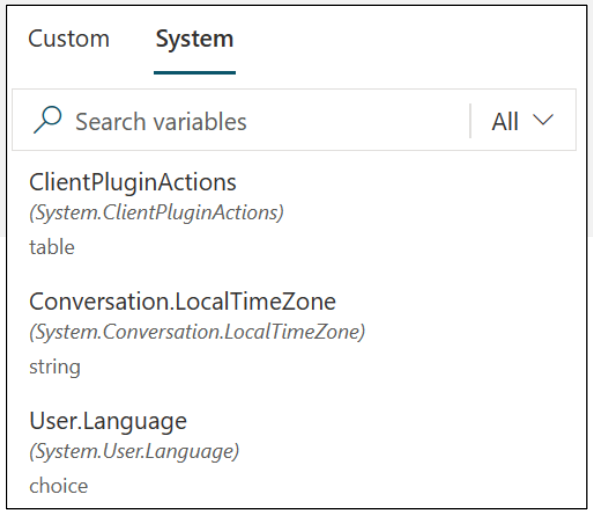
Global variables

These have a broader scope and can be used across all topics. You can convert a topic variable into a global variable.



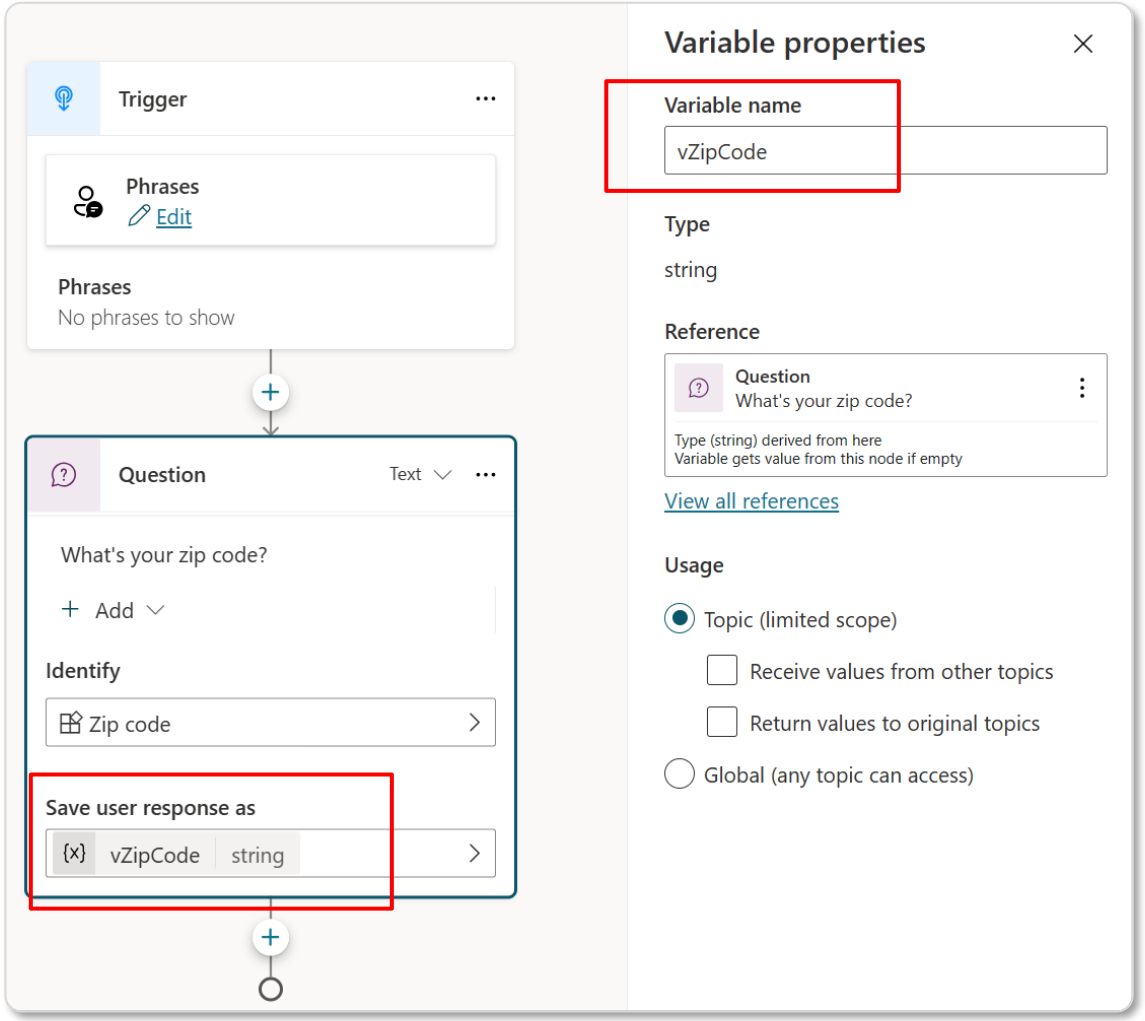
System variables

These are automatically generated with your agent and provide additional context about the conversation or user. They are accessible in all topics.

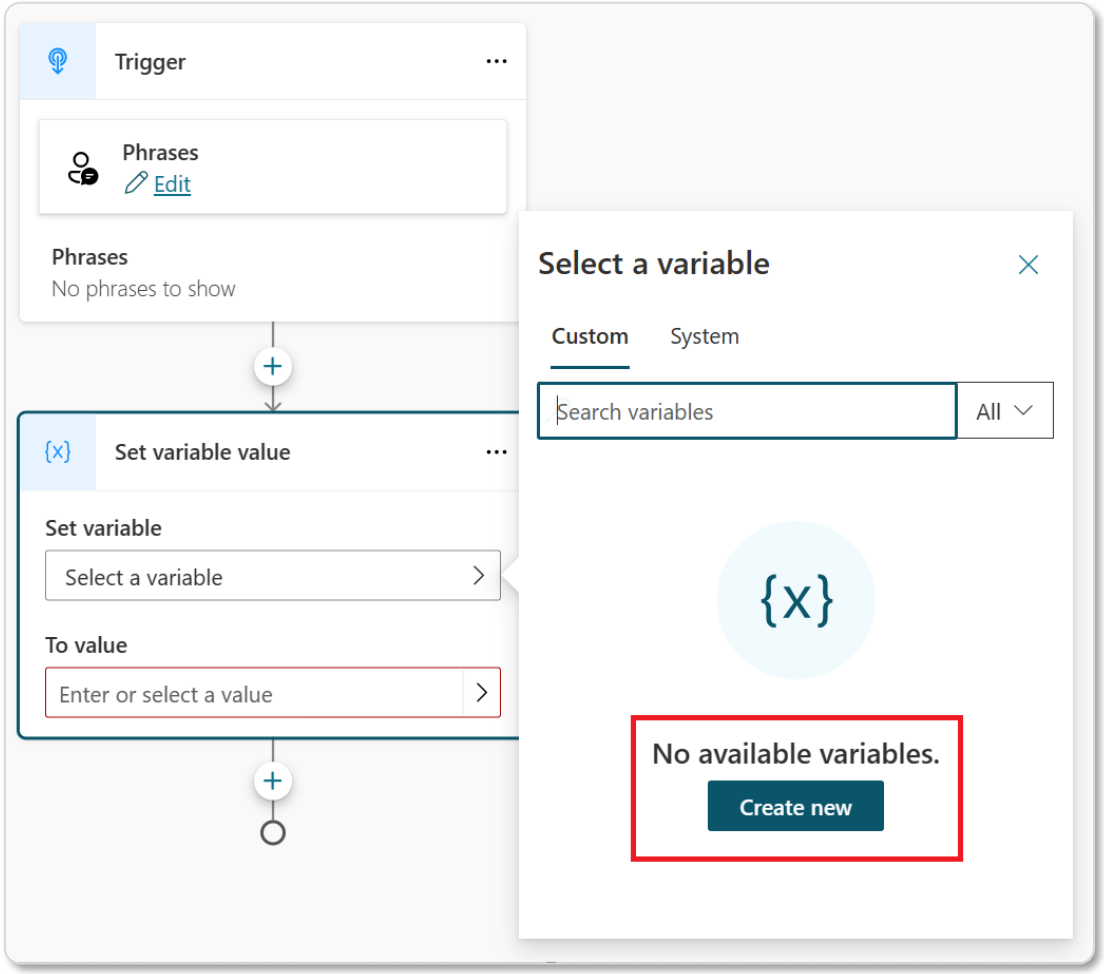


Create variable

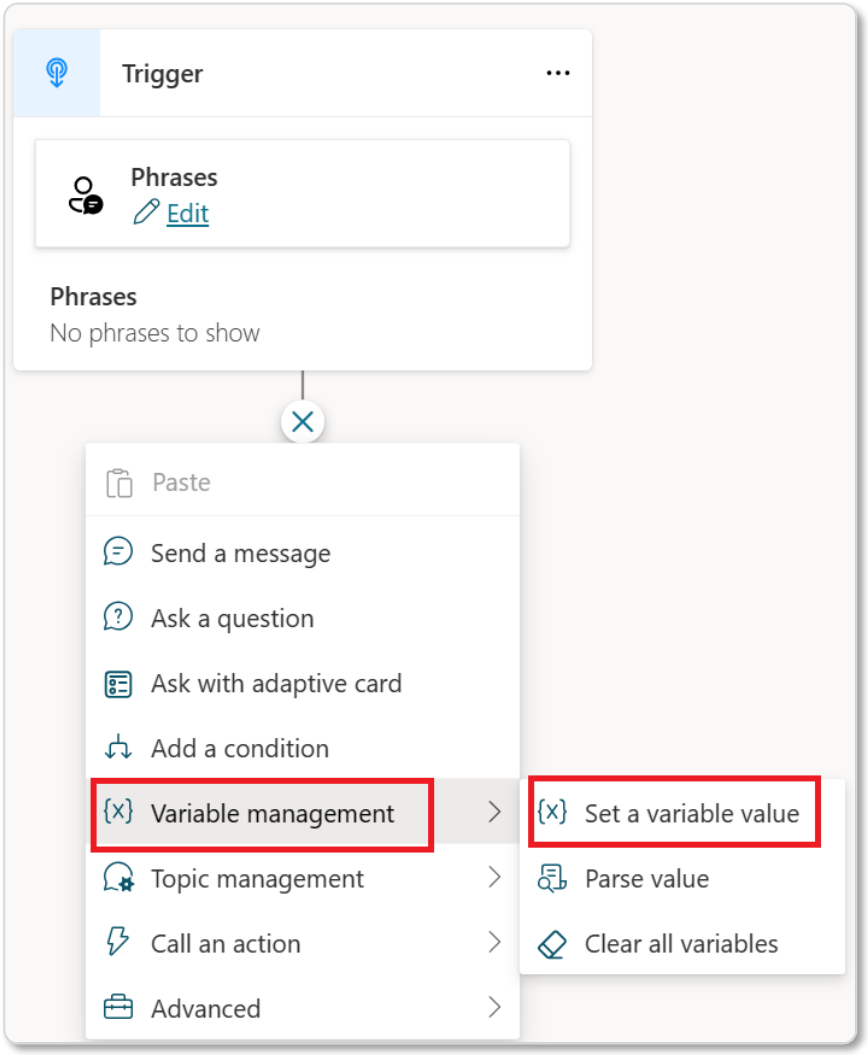
From other nodes. Like 'Question'

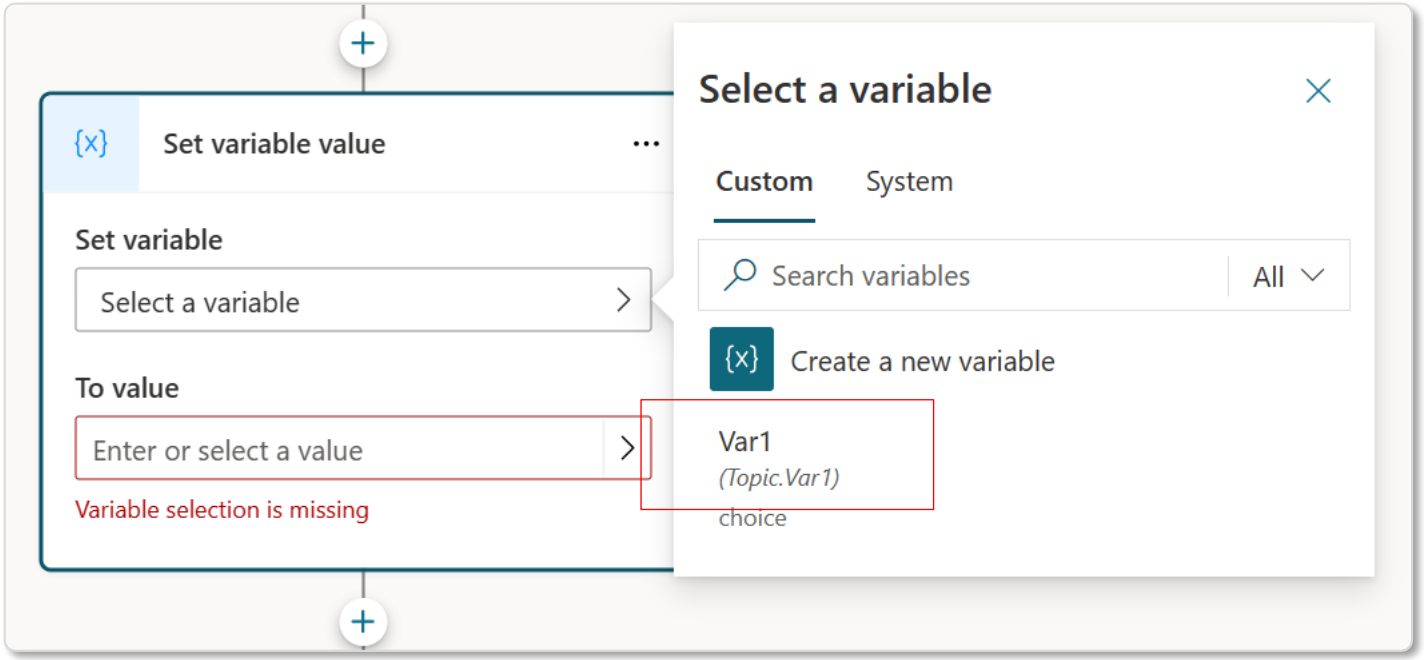


From within set variable node:



Set variable value





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Call HTTP Action & Power automate flow in Copilot Studio

Call a Power Automate flow

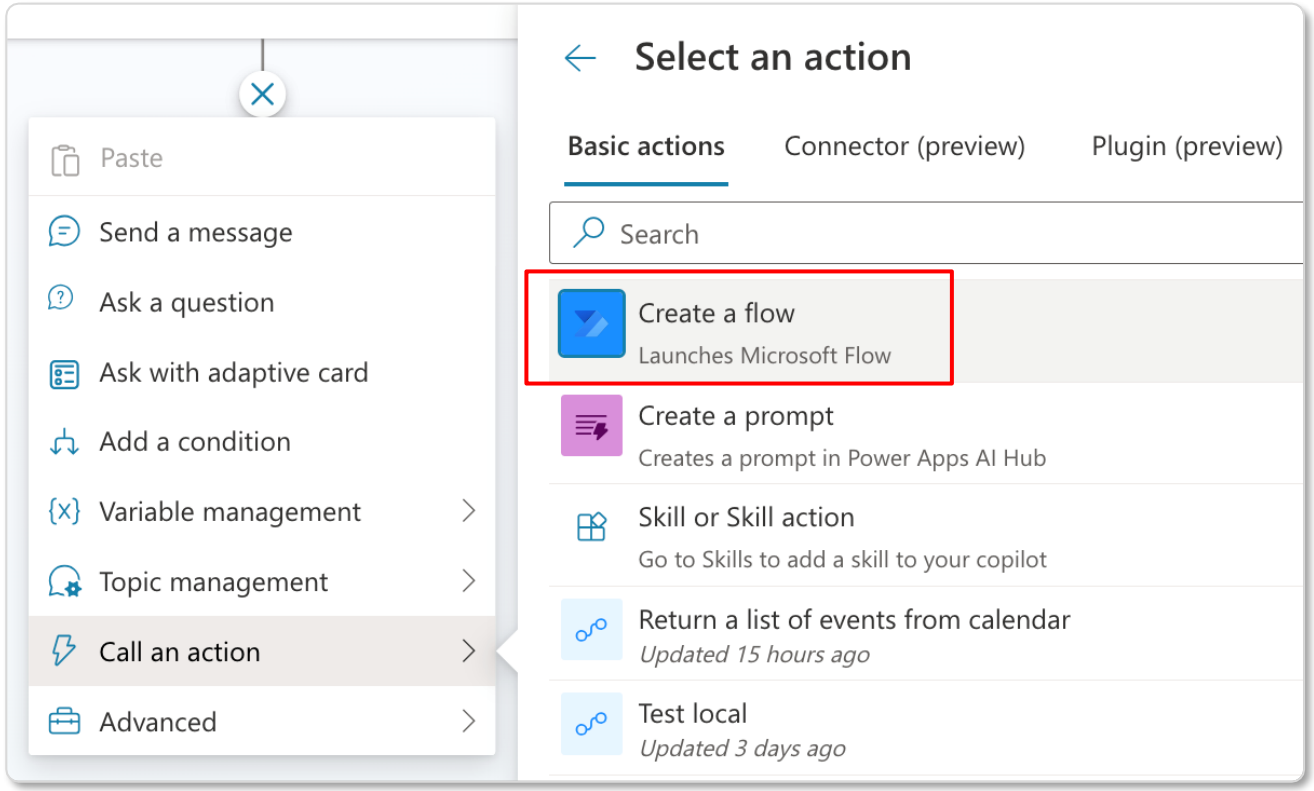
Extend the capabilities of your agent with **cloud flows that you build in Power Automate** using low-code, drag-and-drop tools.

You can use flows that **already exist in your Power Apps environment** or create a flow from the Copilot Studio authoring canvas.

Flows must be part of a solution package with Agent to be available as **generative actions**

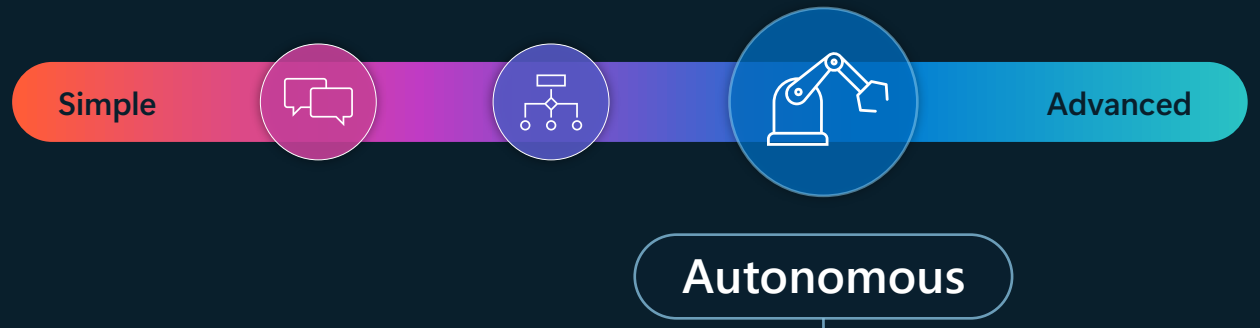
To use cloud flows in custom agents in Copilot Studio, the flows must:

- Have the **Run a flow from Copilot** trigger.
- Flow must be in a solution within the **same environment** as the agent.
- Ensure response action is set to return in real time
- To avoid timeouts, flow response must be completed within the **100-second action limit**.
- **Actions that need to run longer** can be placed after the Respond to Copilot action (duration up to 28 days).



A spectrum of agents

Agents vary in levels of complexity and capabilities depending on your need



What are autonomous agents?

Expert systems that **manage and automate complex business tasks**, enhancing efficiency and innovation across your organization.

Proactive

Automatically responds to signals across your business and initiate tasks, configured to react to events or triggers without human input.

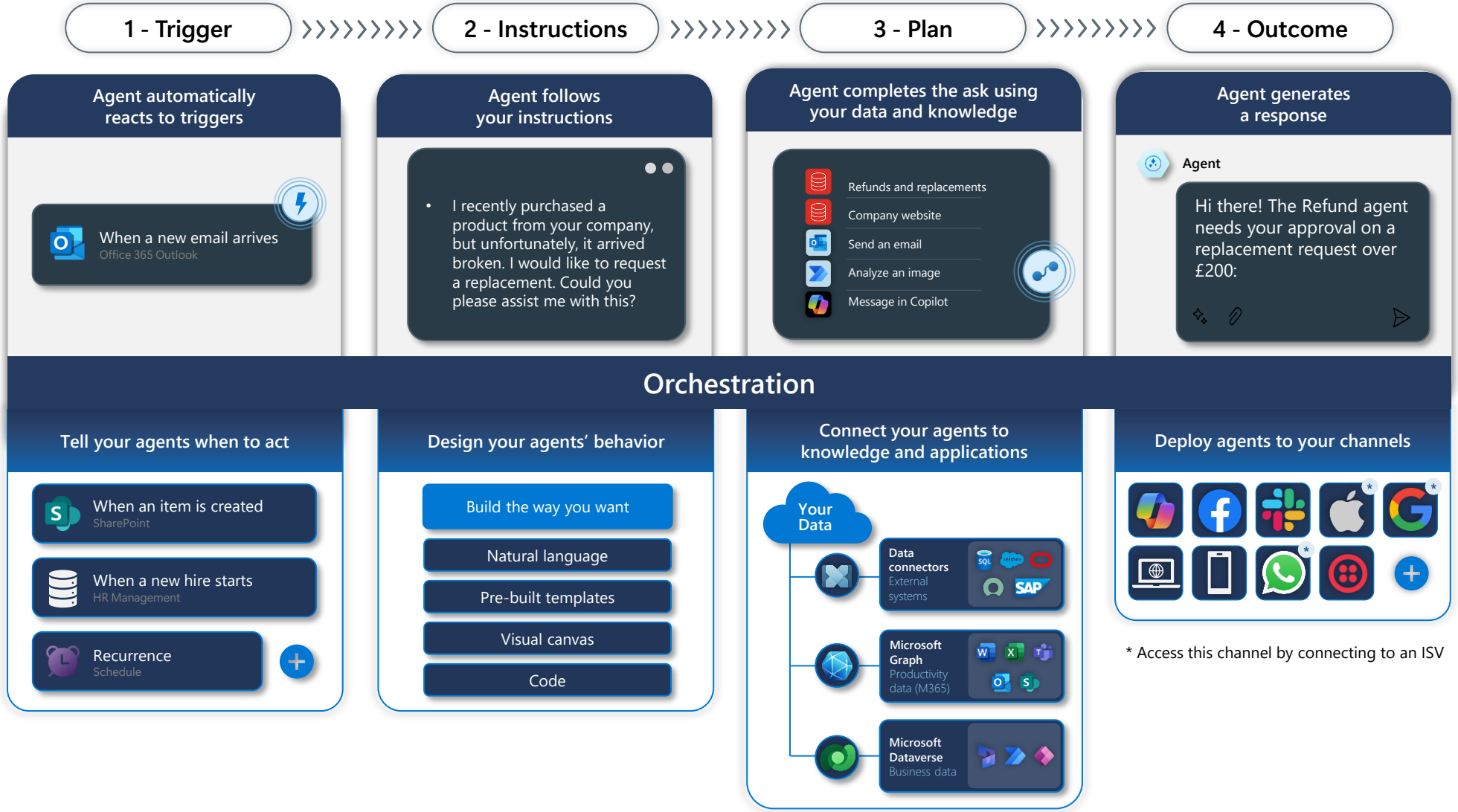
Independent

Seamlessly completes tasks behind the scenes without human intervention.

Adaptive

Non-deterministic and able to produce different outcomes from the same initial conditions.

How does it all fit together?



* Access this channel by connecting to an ISV

Q&A

Thank you